

```
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def is_safe(vertex, color, graph, colors):
```

```
    for i in range(len(graph)):
        if graph[vertex][i] == 1 and colors[i] == color:
            return False

    return True
```

```
def graph_coloring(graph, m, colors, vertex):
```

```
    if vertex == len(graph):
        return True

    for color in range(1, m + 1):
        if is_safe(vertex, color, graph, colors):
            colors[vertex] = color

            if graph_coloring(graph, m, colors, vertex + 1):
                return True

            colors[vertex] = 0

    return False
```

```
graph = [
    [0, 1, 1, 1],
    [1, 0, 1, 0],
    [1, 1, 0, 1],
    [1, 0, 1, 0]
]
```

```
m = int(input("Enter number of colors: "))
```

```
colors = [0] * len(graph)
```

```
if graph_coloring(graph, m, colors, 0):
```

```
    print("Color assignment:")
    for i in range(len(colors)):
        print("Vertex", i, "=", "Color", colors[i])
```

```
else:
    print("Solution does not exist")
```

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Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2025)
on win32

>>> Enter "help" below or click "Help" above for more

==== RESTART: C:/Users/ROOPA SAI DIVYA/OneDrive/Desktop/

Enter number of colors: 3

Color assignment:

Vertex 0 = Color 1

Vertex 1 = Color 2

Vertex 2 = Color 3

Vertex 3 = Color 2

>>>